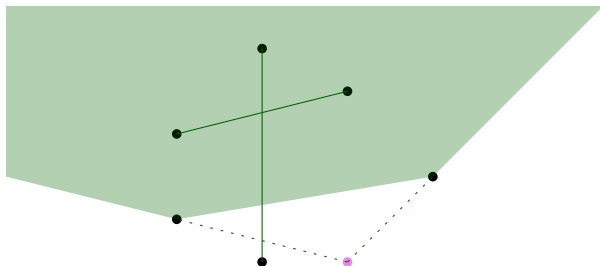


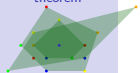
Linear versions of Tverberg's theorem: Progress and problems



Pablo Soberón
Northeastern University
p.soberonbravo@northeastern.edu
September 7th, 2017 - MSRI

Linear versions of

Tverberg's
theorem



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Tverberg's
theorem

Warm-up: two
hard problems

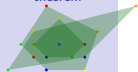
Sarkaria's
technique

Variations

When Sarkaria
fails

(p, q) theorems

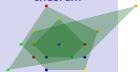
Tverberg's theorem



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Theorem (Tverberg 1966)

Given $(r - 1)(d + 1) + 1$ points in $\mathbb{R}^d$, there is a partition of them into r pieces so that the convex hulls of the parts intersect.



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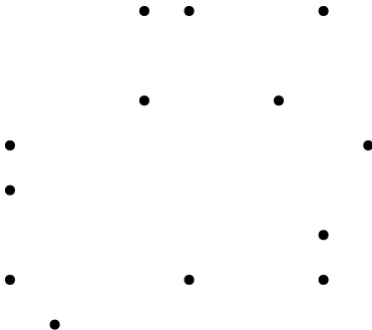
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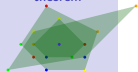
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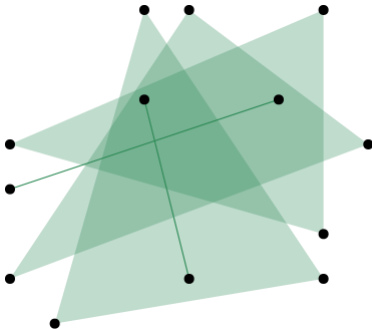




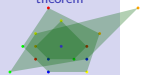
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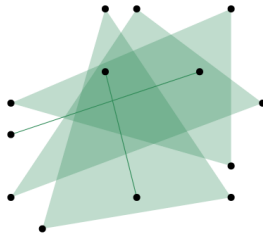
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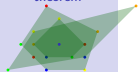
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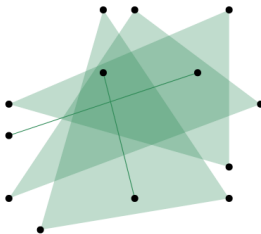
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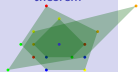


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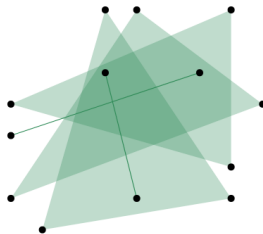


There are several proofs methods to approach this kind of results.

- (Tverberg) Move points continuously

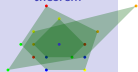


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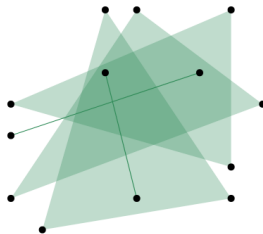


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- (Tverberg) Move points continuously
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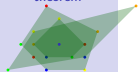


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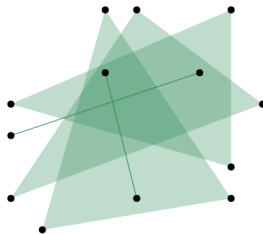


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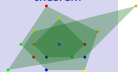


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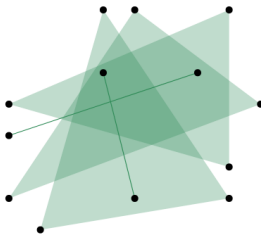


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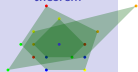


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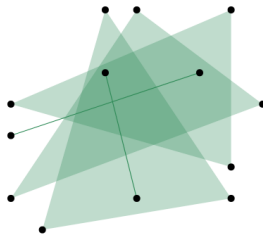


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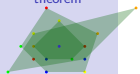


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Goal of the talk: Go over Sarkaria's method and results that can be obtained by modifying it. Then, talk about results which look very different.

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Tverberg's
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Warm-up: two
hard problems

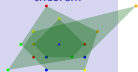
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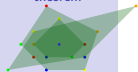
Dutch cheese conjecture



Dutch cheese conjecture

Conjecture (Sierksma)

Given $(r - 1)(d + 1) + 1$ points in $\mathbb{R}^d$, there are at least $(r - 1)!^d$ different Tverberg partitions.

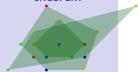


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Examples:



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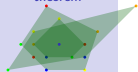
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Examples:

- Clusters around simplex.





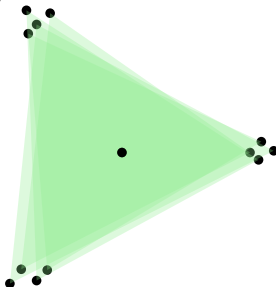
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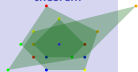
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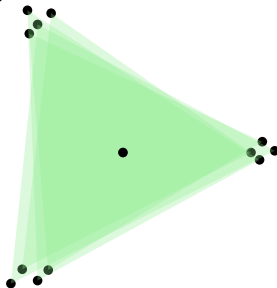
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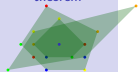
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Examples:

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- Diagonal of “stretched grid” [Bukh, Loh, Nivasch 2017+].



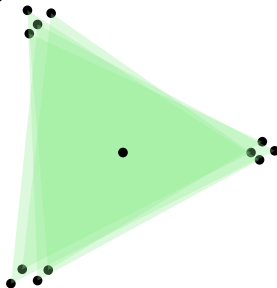
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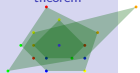
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- Diagonal of “stretched grid” [Bukh, Loh, Nivasch 2017+].
- Only known non-trivial case is $r = 3, d = 2$ [Hell 2008]

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Warm-up: two
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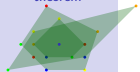
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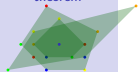


Larman's problem

Conjecture (Larman)

For each d , there is a set of $2d + 2$ points in $\mathbb{R}^d$ so that the following holds. For any partition of them into two sets A, B there is a point x in the set so that

$$\text{conv}(A \setminus \{x\}) \cap \text{conv}(B \setminus \{x\}) = \emptyset$$



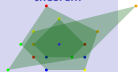
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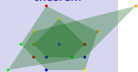
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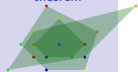
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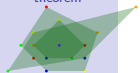
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Warm-up: two
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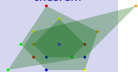
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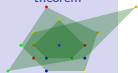
Proving Tverberg's theorem



Proving Tverberg's theorem

If you are confronted with a Tverberg-type theorem, what ideas are useful?

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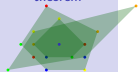
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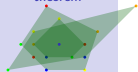
Tverberg's brother: Colorful Carathéodory



Tverberg's brother: Colorful Carathéodory

Theorem (Bárány 1982)

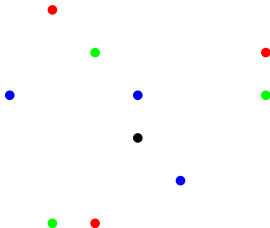
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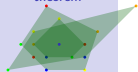


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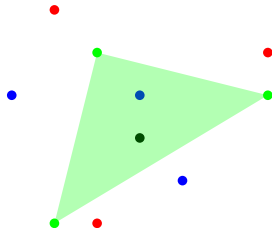


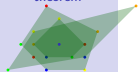


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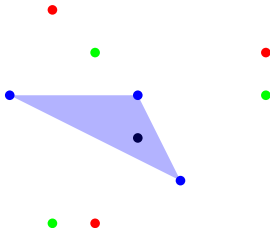


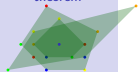


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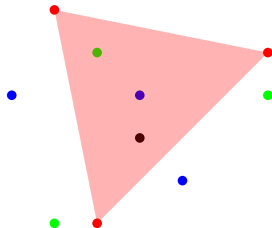


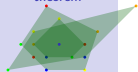


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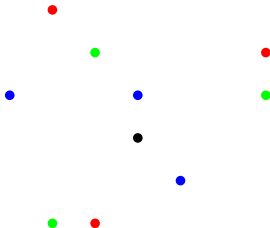


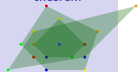


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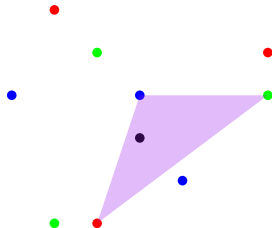


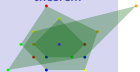


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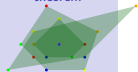




Linear Borsuk-Ulam

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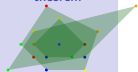


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If we have $|X_i| = 2$ for all i , we may assume without loss of generality that the origin is the midpoint of the segment $\text{conv } X_i$.



Linear Borsuk-Ulam

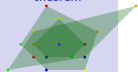
Theorem (Bárány 1982)

Given $d + 1$ sets $X_1, \dots, X_{d+1}$ whose convex hulls contain the origin, there is a colorful choice $x_1 \in X_1, \dots, x_{d+1} \in X_{d+1}$ such that the convex hull of $\{x_1, \dots, x_{d+1}\}$ contains the origin.

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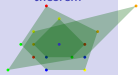
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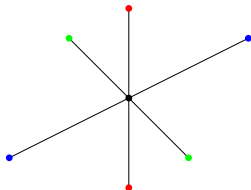
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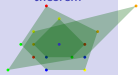
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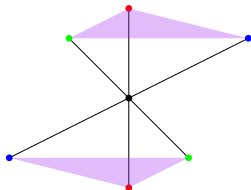
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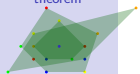
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Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

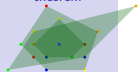
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Variations

When Sarkaria
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(p, q) theorems

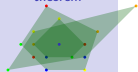
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Sarkaria's transformation

Theorem (Radon 1921)

Given $d + 2$ points in $\mathbb{R}^d$, there is a partition of them into two parts A, B such that their convex hulls intersect.

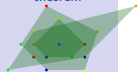


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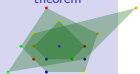
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How do we extend those ideas to general Tverberg partitions?

Linear versions of

Tverberg's
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Tverberg's
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Warm-up: two
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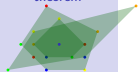
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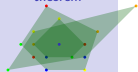
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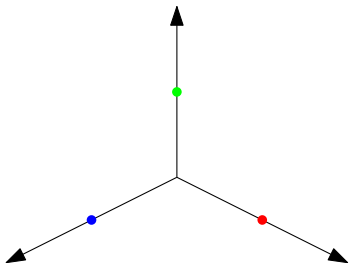
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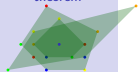
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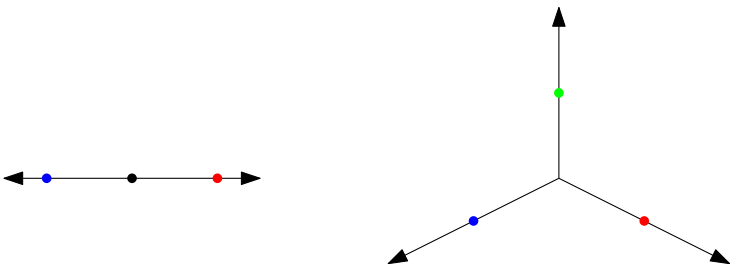
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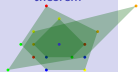


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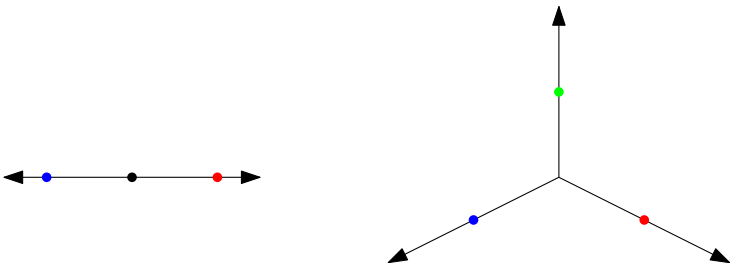


Consider $u_1, \dots, u_r$ as the vertices of a regular simplex in $\mathbb{R}^{r-1}$ centered at the origin.



Sarkaria's transformation

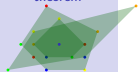
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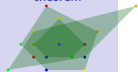
Given a point $a \in \mathbb{R}^d$, consider the points constructed as follows:

$$a \mapsto (a, 1) \otimes u_j \in \mathbb{R}^{(d+1)(r-1)} \text{ for } j = 1, \dots, r.$$



Sarkaria's transformation

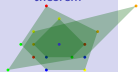
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Sarkaria's transformation

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Let's see what happens if $r = 2$



Pablo Soberón

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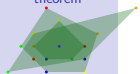
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Linear versions of

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Warm-up: two
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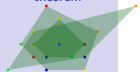
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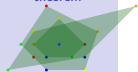
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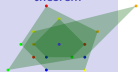


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$a_1, \dots, a_n \longleftrightarrow$

$(a_1, 1) \otimes u_1$	$\dots$	$(a_n, 1) \otimes u_1$
$(a_1, 1) \otimes u_2$	$\dots$	$(a_n, 1) \otimes u_2$
$\vdots$		$\vdots$
$(a_1, 1) \otimes u_r$	$\dots$	$(a_n, 1) \otimes u_r$

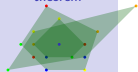


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Tverberg partitions in r parts in $\mathbb{R}^d$ correspond one-to-one to colorful Carathéodory choices in $\mathbb{R}^{(d+1)(r-1)}$.



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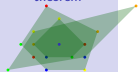
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We can obtain variations of Tverberg's theorem if we modify the transformation.

Linear versions of

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Pablo Soberón

Tverberg's
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Warm-up: two
hard problems

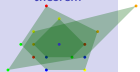
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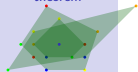
(p, q) theorems

Example 1: Tverberg plus minus



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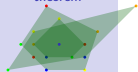
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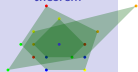
Then,

$$\bigcap_{j=1}^r \text{aff}(A_j) = \{p\}$$

for some $p \in \mathbb{R}^d$.

Linear versions of

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Tverberg's
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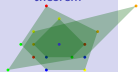
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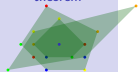
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Theorem (Tverberg plus minus, Bárány, S.- 17+)

Let S be a set of $(r - 1)(d + 1) + 1$ points in $\mathbb{R}^d$ in sufficiently general position. Let $M \subset S$ be a subset of at most $r - 1$ points such that $\text{conv } M \cap \text{conv}(S \setminus M) = \emptyset$.



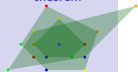
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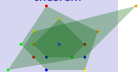
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$M = \emptyset$ is Tverberg's theorem.

Linear versions of

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Tverberg's
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Warm-up: two
hard problems

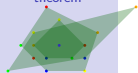
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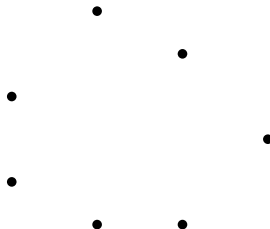
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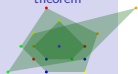
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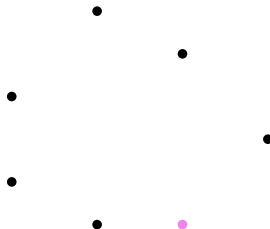
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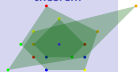
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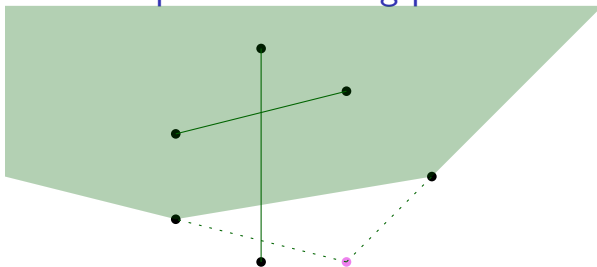
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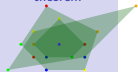
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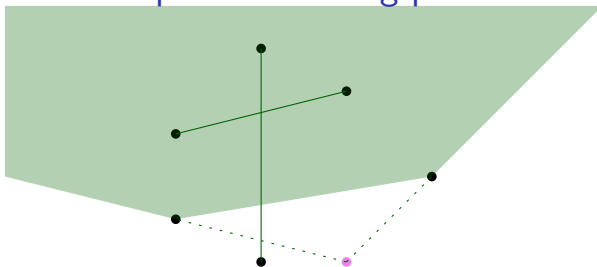
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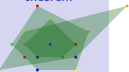


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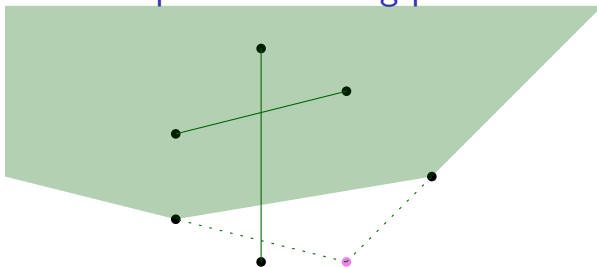


Proof: In Sarkaria's transformation, if $a_i \in M$, consider instead

$$a_i \mapsto (-a_i, -1) \otimes u_j \text{ for } j = 1, \dots, r$$



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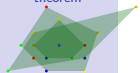


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Problem

Find all integers k such that for any $(r-1)(d+1)+1$ points in $\mathbb{R}^d$ in sufficiently general position, there is a partition of them into r parts whose affine hulls intersect and in the affine combinations that witness the point of intersection there are exactly k negative coefficients.



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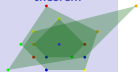
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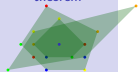
Example 2: Tverberg with tolerance



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Theorem (Bárány 1982)

Given $d + 1$ sets $X_1, \dots, X_{d+1}$ in $\mathbb{R}^d$ whose convex hulls contain the origin, there is a colorful choice $x_1 \in X_1, \dots, x_{d+1} \in X_{d+1}$ such that the convex hull of $\{x_1, \dots, x_{d+1}\}$ contains the origin.

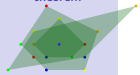


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What if we have more color classes?



Example 2: Tverberg with tolerance

Theorem (Bárány 1982)

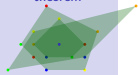
Given $d + 1$ sets $X_1, \dots, X_{d+1}$ in $\mathbb{R}^d$ whose convex hulls contain the origin, there is a colorful choice $x_1 \in X_1, \dots, x_{d+1} \in X_{d+1}$ such that the convex hull of $\{x_1, \dots, x_{d+1}\}$ contains the origin.

What if we have more color classes?

Theorem (S.- 17+)

Given N sets $X_1, \dots, X_N$ of r points each whose convex hulls contain the origin, there is a colorful choice $x_1 \in X_1, \dots, x_N \in X_N$ such that every half-space that contains the origin also contains at least

$$\frac{N}{r} - \sqrt{\frac{dN \ln(Nr)}{2}} \text{ points of } \{x_1, \dots, x_N\}.$$



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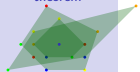
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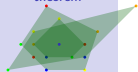
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Proof: Make a random choice, use the probabilistic method.



Example 2: Tverberg with tolerance

We have a colorful Carathéodory that resists the removal of points. If we “process it” through Sarkaria’s machinery, we get the following.



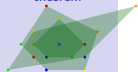
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Given r, t, d , there is an integer $N = rt + \tilde{O}(r^2\sqrt{td} + r^3d)$ such that among any N points in $\mathbb{R}^d$, there is a partition of them into r sets $A_1, \dots, A_r$ such that for every subset C of t of the points, we have

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Example 2: Tverberg with tolerance

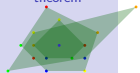
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Previous versions with $N = rt + o(t)$ were known before, using geometric Ramsey-type results. [García-Colín, Raggi, Roldán-Pensado 2017+]



Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

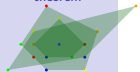
Sarkaria's
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Variations

When Sarkaria
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(p, q) theorems

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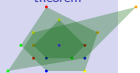
Example 2: Tverberg with tolerance

Problem

Prove a non-trivial Topological version of Tverberg with tolerance.

Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
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Warm-up: two
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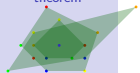
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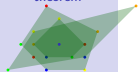
(p, q) theorems

Example 3: colored Tverberg



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Suppose we are given families of r points each instead of points.

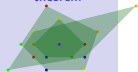


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Theorem (Col. Tverberg with equal coefficients, S.- 2013)

*Given $(r - 1)d + 1$ families $\mathcal{F}_1, \dots, \mathcal{F}_{(r-1)d+1}$ of r points each in $\mathbb{R}^d$ (considered as color classes), there is a **colorful** partition of them into r sets $A_1, \dots, A_r$*

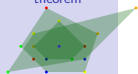


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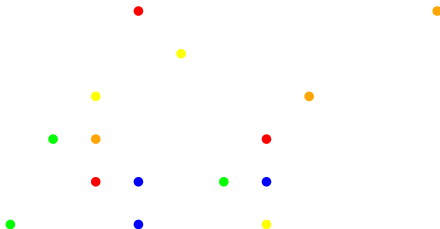


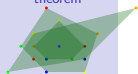
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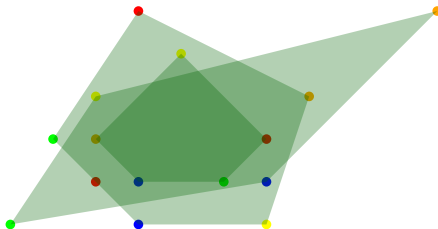


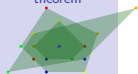
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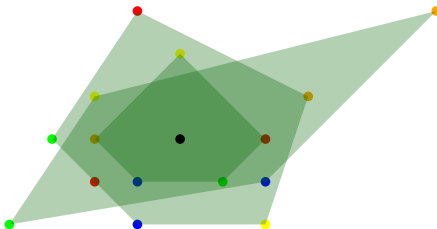


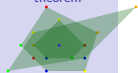
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Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

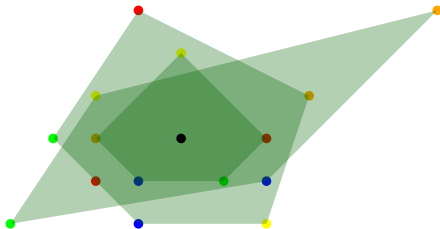
Sarkaria's
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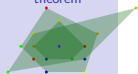
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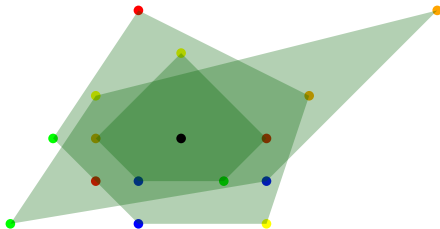
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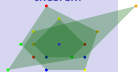
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Proof:



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Tverberg's
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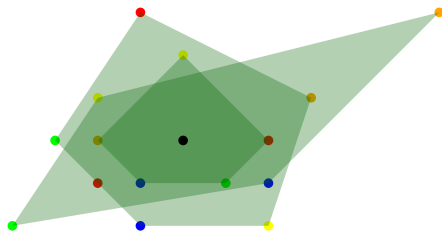
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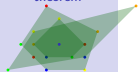
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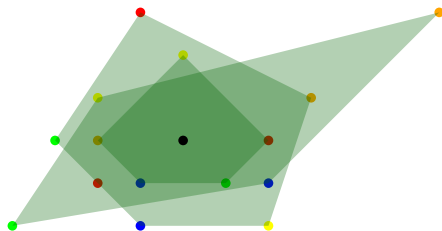
Proof: Given $\mathcal{F} = \{z_1, \dots, z_r\}$ we can consider

$$\mathcal{F} \mapsto \sum_{j=1}^r z_j \otimes u_{i+t} \in \mathbb{R}^{d(r-1)} \quad \text{for } t = 1, 2, \dots, r$$





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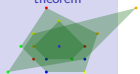
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Topological version in [Blagojević, Frick, Ziegler 2014]

Linear versions of
Tverberg's
theorem



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Tverberg's
theorem

Warm-up: two
hard problems

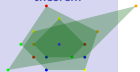
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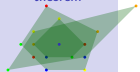
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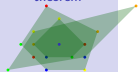
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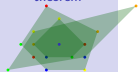
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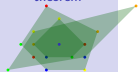
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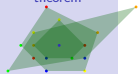


Sarkaria's technique

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- There are many ways to modify it.
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- It's a good source of topological problems.

Linear versions of
Tverberg's
theorem



Pablo Soberón

Tverberg's
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Warm-up: two
hard problems

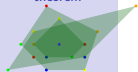
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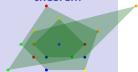
Versions where Sarkaria's technique fails



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Theorem (Integer Radon, Bezdek, Blokhuis 2003)

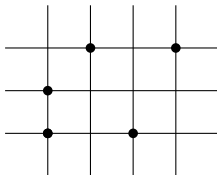
Given 17 points with integer coordinates in $\mathbb{R}^3$, there is a partition of them into two sets A, B such that their convex hulls share a point with integer coordinates.

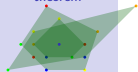


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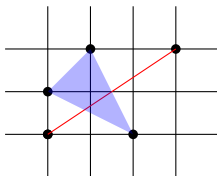


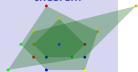


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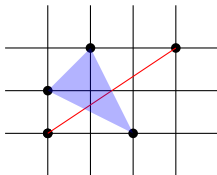




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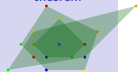


Problem

The result above fails if we use 10 points or less. What is the minimum number of integer points to guarantee such a Radon partition?

Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

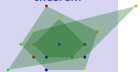
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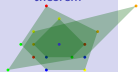
(p, q) theorems

Integer Tverberg



Integer Tverberg

Integer Tverberg can be naturally extended, and we can expect more.

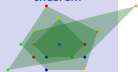


Integer Tverberg

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Definition

Let $ZT(r, d)$ be the smallest integer such that among any $ZT(r, d)$ points of $\mathbb{Z}^d$ there is a partition of them into r parts $A_1, \dots, A_r$ such that their convex hulls share a point of $\mathbb{Z}^d$.



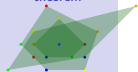
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It's known that $(r-1)2^d + 1 \leq ZT(r, d) \leq (r-1)d2^d + 1$.



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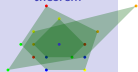
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Problem

Determine the asymptotic behavior of $ZT(r, d)$. Is it closer to $r2^d$ or to $rd2^d$?



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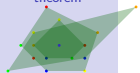
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What if instead we want the intersection of the convex hulls to be large?

Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

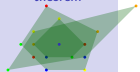
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Diameter Tverberg

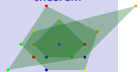


Diameter Tverberg

Theorem (Tverberg)

Let r, d be positive integers. Given $(r - 1)(d + 1) + 1$ points in $\mathbb{R}^d$ there is a partition of them such that

$$\bigcap_{j=1}^r \text{conv}(A_j) \neq \emptyset.$$

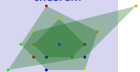


Diameter Tverberg

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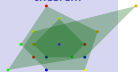


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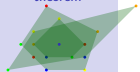


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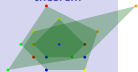


Diameter Tverberg

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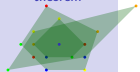


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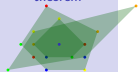


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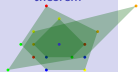


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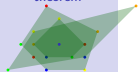


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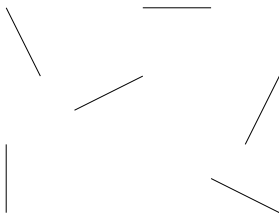


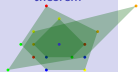
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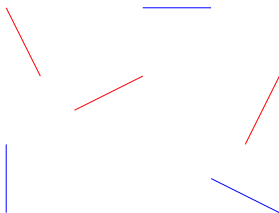


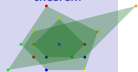
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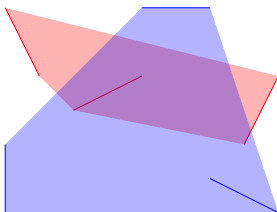


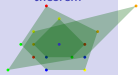
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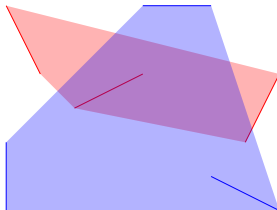


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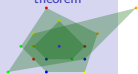


Problem

For the volume?

Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
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Warm-up: two
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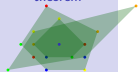
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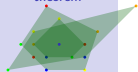
(p, q) theorems

Quantitative Tverberg theorems



Quantitative Tverberg theorems

Suppose we have a way of deciding if a convex set is “large”.



Quantitative Tverberg theorems

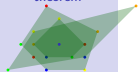
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Given N large convex sets in $\mathbb{R}^d$, when is it possible to split them into r families $A_1, \dots, A_r$ such that

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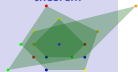
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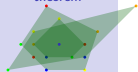
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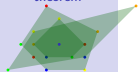
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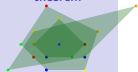
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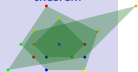
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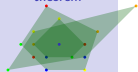
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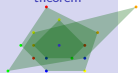
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- Has k integer points
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- Has large volume
- Has large surface area

Linear versions of

Tverberg's
theorem



Pablo Soberón

Tverberg's
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Warm-up: two
hard problems

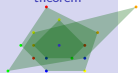
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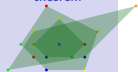
(p, q) theorems

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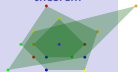


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Theorem (Alon, Kleitman 1992)

Given $p \geq q \geq d + 1$, there is a constant $c = c(p, q, d)$ such that the following holds. For any finite family $\mathcal{F}$ of convex sets such that out of every p of them there are q which intersect, there are c points whose union intersects every set in $\mathcal{F}$.

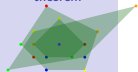


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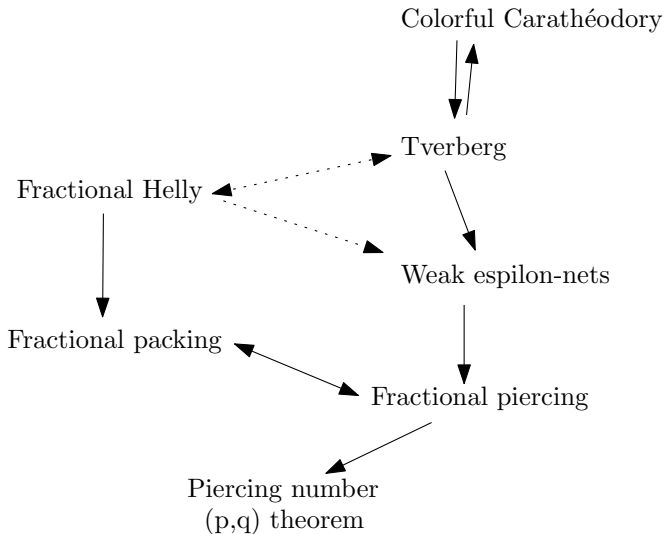
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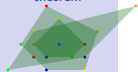
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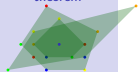
Proof of (p, q)





An application

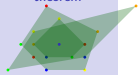
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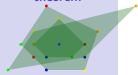
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Given a finite family $\mathcal{F}$ of at least $2d$ non-empty convex sets in $\mathbb{R}^d$, if the intersection of any $2d$ of them is not empty and has diameter at least 1, then $\cap \mathcal{F}$ is not empty and has diameter at least d^{-2d} .



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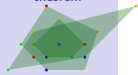
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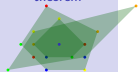
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Theorem (De Loera, La Haye, Rolnick, S.- 2017)

Given $\varepsilon > 0$, there is an integer $N(\varepsilon) = \Theta_d(\varepsilon^{-(d-1)/2})$ such that the following holds. For any finite family $\mathcal{F}$ of non-empty convex sets in $\mathbb{R}^d$, if the intersection of any $N(\varepsilon)$ of them is not empty and has diameter at least 1, then $\cap \mathcal{F}$ is not empty and has diameter at least $1 - \varepsilon$.



Pablo Soberón

Tverberg's
theorem

Warm-up: two
hard problems

Sarkaria's
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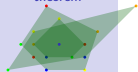
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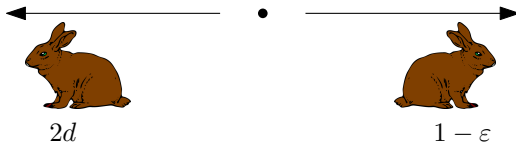
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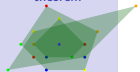
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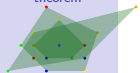




Two hares

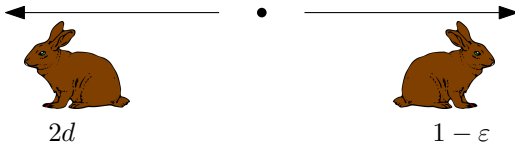
Theorem

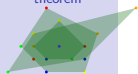
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Pablo Soberón

Two hares





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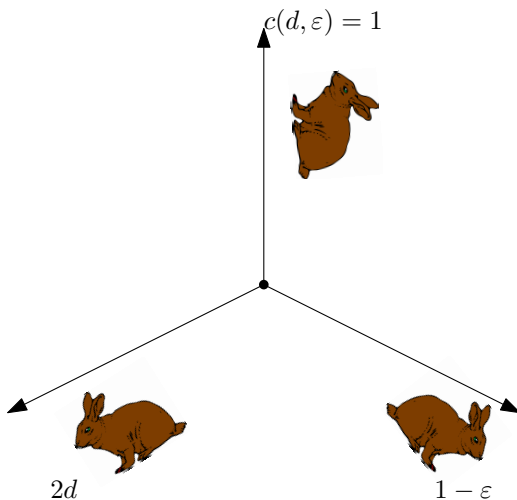
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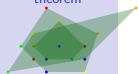
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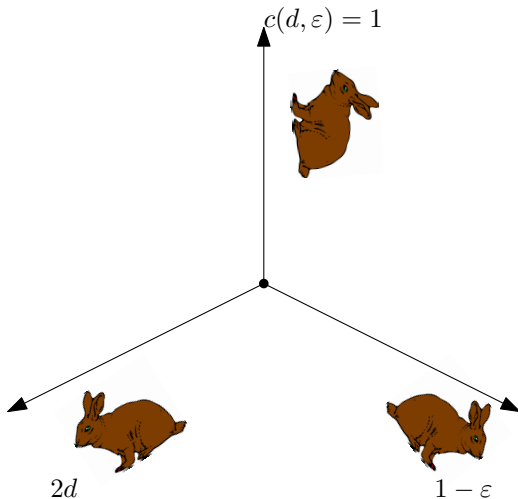
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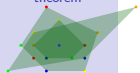


Problem

For the volume?

Linear versions of

Tverberg's
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Thanks!